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PHYSICAL PHARMACEUTICS -I

UNIT - 4th

COMPLEXATION AND PROTEIN BINDING

- Complexation is defined as an association of two or more chemical species, resulting a formation of complexes.
- Complexes generally result from a donor-acceptor mechanism.

Eg: NH_3 (Donor → Ligands (which has lone pair)).

(NH_3) (Donor Compound)

- The Donor compound is a non-metallic atom or ion, which can donate an electron pair.
- The acceptor is usually a metallic ion(atom) or a neutral atom is capable of accepting a pair of electrons. eg. Co (cobalt).

So,
Subspait (Acceptor) molecule referred as central atom.
and Donor referred as ligands, which attached to the central atom.

- Classifications:-

1) Metal ion Complexes / Coordinate Complexes
   - Inorganic type
   - Chelates
   - Olefin types
   - Aromatic types.
2) Organic Molecular complexes
   - Quinhydrone type
   - Picric acid type
   - Caffeine and other drug Complexes
   - Polymer type
3) Inclusion / Occlusion Compounds.
   - Clathrate
   - channel lattice type
   - layer type
   - Macromolecular type
   - Macromolecular type

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- Donor (Ligands) :— which has lone pairs and attached with central atom.

Types of ligands :—

i) Monodentate → which provide only one centre for attachment to the central atom.

Eg. NH_3 , H_2O , Cl^-

ii) Bidentate ligand → Those ligands which provide two centre for attachment to the central metal ion(atom).

Eg. $H_2N-CH_2-CH_2-NH_2$ [Ethylenediamine]

iii) Polydentate ligand → Those ligand which provide two or more than two centre for attachement to the central atom. (Hexadentate)

Eg.

[EDTA]

- Applications of Complexation

i) Physical state → To convert liquid substances to solid complexes and improve its processing characteristics.

ii) Stability of drugs → Stability of drugs can be enhanced by complexation.

iii) Solubility → Enhancement of solubility by using solid complexes.

iv) Dissolution → Complexation enhances the solubility, thus enhancing the dissolution of drugs.

v) Absorption & Bioavailibility → Complexation helps in increasing the absorption and bioavailability of drug in the body.

vi) Antidote for metal poising → Also reduced toxicity of poisons, by making complex with metal poison. So, work as antidote.

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1) Metal ion Complexes / Coordinate Complex

In this type, metal ion constitutes as central atom and interacts with ligands.

i) Inorganic Complexes :→

In this complexe, ligand are inorganic in nature, which attached with metal atom.

Eg. ligands we → NH_3, H_2O:, $C\equiv N$: (all are inorganic).

① $[Ni(H_2O)_4]Cl_2 \longrightarrow$ $2Cl^- + [Ni(H_2O)_4]^{2+}$

I° = 2
II° = 4
C.N. = 6

In which, ligands attached with metal atom with coordinate bond.
And, Cl attached with simple ionic bond.

• Valencies

→ primary valencies (1°) — Ionic bond
→ secondary valencies (2°) — Coordinate bond or total no. of ligands attached to central atom.
→ Coordination number — maximum number of atom binding to the central atom.

Eg. ④ $[Co(NH_3)_6]^{+}$

C.N. = 6

ii) Chelates →

A substance containing two or more donor group may combine with a metal ion to form a complex known as chelate.

→ Bonds involes may be → ionic, primary covalent type or coordinate covalent type.

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→ Depending on ligands group, it may be bidentate, tridentate, hexadentate or polydentate.

→ When metal atom (central) attached with these ligands, it form cyclic structure (closed).

Eg ⇒ EDTA, ethylenediamine etc--

iii) Olefin complex → Aqueous solution of certain metal ion such as platinum, ion, palladium, silver etc. can absorb olefins such as ethylene to yield water soluble complexes.

Eg. Silver-Olefin Complexes

iv) Aromatic Complexes

They are formed by interaction of metal ions as acceptors with aromatic molecules such as benzene, toluene & Xylene

→ π bond
→ σ bond
→ Sandwich Compound

• π bond ⇒

• σ bond ⇒

• Sigma-bond complex of toluene with HCl·AlCl3

• Sandwich compound ⇒ Sandwich compounds are relatively stable complexes involving a delocalized covalent bond between the d-orbital of a transition metal and a molecular orbital of aromatic ring.

(ferrocene)

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ii) Organic Molecular Complexes :

Organic molecular complexes, also known as addition complexes are formed by the union of two organic molecule held together by electrostatic forces, ionic, covalent and also by hydrogen bonded complexes.

- Charge transfer Complexes →

These complexes are usually formed by sharing of π-electrons. In these types of complexation one of the constituent molecules of the complex polarizes the other resulting in a type of ionic interaction or charge transfer.

(charge-transfer complex between benzene and tri-nitro benzene)

iii) Quinhydrone Complex (types) →

The molecular complex of this type is obtained by mixing alcoholic solutions of equimolar quantities of benzoquinone and hydroquinone

iv) Picric acid complexes →

Picric acid (2,4,6-trinitrophenol), being a strong acids, forms organic molecular complexes with weak bases.

- examples → Butesin picrate

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Caffeine and other drug molecules →

Caffeine forms complexes with a number of drugs such as benzocaine, tetracaine or procaine and this enhances the stability and appearance of pharmaceuticals preparation of these drugs.

Polymer type Complex →

Many pharmaceutical additives such as polyethylene glycols (PEGs), polysorbate, Carboxymethyl cellulose (CMC) etc... can form complexes with a large number of drugs.

→ It involves dipole-dipole forces/hydrogen bonding.

example →

iii) Inclusion Complex / Occlusion compounds

In these complexes, one of the components is entrapped (trapped) in the open lattice or cage-like crystal structure of the other.

- Host component is known as Guest molecules and open lattice and cage like structure is know as host molecule in which guest molecule trapped.
- No involvement of any type of bond, so also called no-bond complexes.

- Clathrates →

Clathrates are inclusion compounds in which a molecules of guest compound get entrapped within the cage like structure found by the association of several molecules of host compounds.

eg

- Hydroquinone in Benzoquinone

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- **Channel lattice type →**

  Channels are formed by crystallization of the host molecules in which the guest molecules can fit, the guest component is usually limited to long, unbranched straight chain compounds.

- **Layer type →**

  In this type, Guest molecules is diffused between the layers of carbon atom, hexagonally oriented to form alternate layers of guest and host molecules.

  Eg → clays, montmorillonite

- **Monomolecular complexes :-**

  In this complexes, a single guest molecules is entrapped in the cavity of host molecules, eg → most commonly used (employed) are :- Cyclodextrin Complexation.

- **Macromolecular Complexes :-**

  In this complexes, more than one guest molecules is entrapped in the cavity of host molecules eg. Cyclodextrin as a host molecules.

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- **Methods of analysis of Complexation**

After complexation process, we have to know whether the complex is made or not?

for this, we use following methods:—

i) Method of continuous variation

ii) Distribution method

iii) Solubility method

iv) pH titration method

i) **Method of Continuous Variation :-**

We know than, when two or more species associated, they formed a complex and due to formation of complex their physical properties would have changed such as Dielectric constant, refractive index etc.-

ii) **Distribution method :-**

In distribution method, we find out partition coefficient (distribution coefficient).

Eg. $K = \frac{X \text{ in Oil}}{X \text{ in water}}$

Now, If, in distribution method complex formed than its solubility increase in oil/water & due to it, its partition coefficient's value change

Let,

① → when complex is not formed

Iodine (I_2) in 30 ml CCl_4 & 30 ml water

for this, let Partition coefficient (K) = a

② → when complex is formed/not formed (check).

Iodine (I_2) in 30 ml CCl_4 & 30 ml KI

for this, let Partition coefficient (K) = b

If complex is formed in 2nd condition,

then $a \neq b$.

If complex is not formed in 2nd condition,

then $a = b$. (same in second as first).

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iii) **Solubility method :-**

In this method, the complex formation is based on the solubility of the components in presence of a complexing agent.

Eg: Complexation of PABA (para-amino benzoic acid) by caffeine

PABA + Caffeine ⟶ PABA-Caffeine complex

$$K = \frac{[PABA\text{-}Caffeine]}{[PABA][Caffeine]}$$

If, K (constant) value increase → Complexation ↑
K (constant) value decrease → Complexation ↓

iv) **pH titration method :-**

When complex formed (complexation happened) then also its pH change,
So, firstly we check pH before complexation, then after complexation if the value of pH change means complex formed and if pH not change means complex not formed.

example → Chelation of cupric ion by glycine

① → Glycine + NaOH ⟶ Basic pH

② → Glycine + NaOH + Cu^{2+} → Acidic pH (means complex formed, pH change)

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Protein binding, Complexation and drug action, Crystalline structure of complexes and thermodynamic treatment of stability constants.

- **Protein Binding :-**

The phenomena of complex formation of drug with protein is called as protein drug binding (protein binding).

eg ⇒

The proteins which involves in this complexation are :- (Blood protein)

- Albumin (mostly drug attached with albumin)
- α-acid glycoprotein
- Lipoprotein
- Globulins

these proteins are present in blood plasma protein.

- their are some drug which bind with protein of blood cells & protein of extravascular tissue
- But in maximum cases, drug bind with blood protein.

- **Mechanism of protein binding**

i) **Reversible →** In this type of protein binding, the drug is bind with the protein with very weak forces like hydrogen bonding, weak vander waal's forces of attraction.

- So, they can easily release the drug and drug becomes free, and it binds with the receptor
It is responsible for the pharmacological action of the drug

ii) **Irreversible →** In this type of protein binding, the drug bind with the protein with strong bond like ~~ionic bond~~ covalent bond.

- Drug after binding with protein cannot release and drug do not become free so they not give any pharmacological action.

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Complexation and drug action

- Complexes can alter the pharmacological activity of the agents by inhibiting interaction with receptor.
- Protein binding complex also affect absorption, Metabolism and drug action by interfering with receptor site of action. (inactivate the drug)
- The action of drug to remove toxic metals in from human bodies is through Complexation reaction.

In some cases, complexation also lead to poor solubility or decrease absorption of the drugs in the body.

The irreversible protein binding also inhibit the drug of action by not release the drug from protein.

Example:- Complexation of drug in the GIT fluids may alter rate and extent of drug absorption.

Crystalline Structure of Complexes

- A crystalline structure is any structure of ions, molecules, or atoms that are held together in a ordered, three-dimensional arrangement.

Eg ⇒

- NaCl

- $[CoCl_2(NH_3)_4]^+$

- Complex compounds cover the range from quite simple inorganic salts to elaborate metal organic hybrid materials

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Their present uses and their potential applications are diverse due to their composition, molecular and crystalline structure and their physical and chemical properties

- **Thermodynamic treatment of stability constants**

The stability constants of the metal complexes are related to thermodynamic product properties such as free energy (ΔG), enthalpy (ΔH) and entropy change (ΔS)

$$\boxed{\Delta G = \Delta H - T\Delta S}$$ → Standard entropy change obtained from this eqn.

where, ΔG = Gibbs free energy
ΔH = Enthalpy
T = Temperature
ΔS = Entropy.

- If ΔG = -ve, then Rate of complexation increases, and if, Rate of complexation increases, then stability constant increases.

(ΔG = -ve) ∝ Rate of complexation ↑ ∝ Stability constant ↑

- If ΔG = +ve, then Rate of complexation decreases, then stability constant decreases.

(ΔG = +ve) ∝ Rate of complexation ↓ ∝ Stability constant ↓

- If
  Temperature ↑ = ΔG ↓ = -ve
  Temperature ↓ = ΔG ↑ = +ve

- **Stability constant** → It is an equilibrium constant for the formation of a complex in solution.
- It is the measure of the strength of the interaction between the reagent that come together to form the complex.

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